

 *Equilux*

Pay, proven equal. No salary revealed.

The first employee-verifiable pay-transparency reporting protocol. A company proves its legally required gender pay gap report was computed from its complete, co-signed payroll — while every salary stays sealed in Midnight's private state.

Midnight Buildathon · Wave 1

Compact · TypeScript · Apache 2.0

github.com/AustinChris1/Equilux

The law now demands a number companies cannot prove.

7 June
2027

FIRST REPORTS DUE • 2026 PAY
DATA

250+

EMPLOYEES, ANNUAL • 150+
SAME DATE, TRIENNIAL

$\geq 5\%$

UNJUSTIFIED GAP IN A
CATEGORY → JOINT PAY
ASSESSMENT

Article 9 requires seven indicators. Article 7 gives every employee the right to ask what their category earns. And today every one of those reports is **self-declared**: nobody can verify the numbers without seeing everyone's salary.

Trust me, or prove it.

Today

HR exports payroll to a spreadsheet, computes a gap, publishes a PDF. An employee cannot check they were counted. A works council cannot check nobody was dropped. A regulator can only audit by subpoenaing the whole payroll — the data the law is trying to protect.

With Equilux

Every employee seals their record. The employer co-signs it. A zero-knowledge circuit computes the gap over the sealed set and publishes only the figure plus a proof that it came from all of it. Verification takes seconds. No salary ever leaves the company.

Equilux does not replace the HRIS. Payroll stays in Personio, DATEV or Workday. Equilux is the **notary on top**: it proves the set was complete and the published figure matches it.

Trust neither side. **Verify both.**

Completeness

The report must cover every enrolled record. A payroll witness with fewer records than were enrolled is rejected — dropping an employee is cryptographically impossible.

No invented records

Every record carries an employee-held secret and an employer attestation. The circuit re-derives each commitment; a forged row fails.

No double counting

Enrollment nullifiers make counting anyone twice unprovable.

An exact figure

The claimed gap is verified in-circuit without division, by cross-multiplied bounds. The circuit accepts one value and rejects every other — one basis point off fails.

Plus an **inclusion receipt** for every employee: a Merkle path proving their record is inside the numbers, revealing nothing about anyone.

Both sides sign. The circuit does the math. Everyone can check.

01 • enroll

Seal

Employee seals salary + gender marker as a commitment and a nullifier. Nothing else touches the chain.

02 • attest

Co-sign

The employer, gated by its secret key, attests each commitment it recognizes from payroll.

03 • publishReport

Prove

Over the full witness: every record enrolled and attested, pairwise distinct, count equals enrolled, gap claim verified. Discloses only the report.

04 • checkReceipt

Verify

Any employee proves their leaf is in the commitment tree against the on-chain root.

Private state holds the payroll. Public state holds the proof.

Private state • witnesses

- salary, gender marker per employee
- employee enrollment secret
- employer secret key
- the full 16-slot payroll witness
- the claimed gap (verified, never trusted)

Public ledger

- commitments — a depth-5 Merkle tree
- nullifier set, attested set, enrolled counter
- employer public key (sealed)
- `PayReport` : headcounts, meanGapAt0rAbove5pct
mean gap in
bps, direction,
- the zero-knowledge proof for every transition

Every witness value that reaches the ledger passes an explicit `disclose()`. The compiler EQUILUX refuses anything else.

Compiled. Tested. Run end-to-end on a Midnight network.

4

CIRCUITS COMPILING · PROVER
+ VERIFIER KEYS

21

TESTS PASSING · CIRCUIT-
LEVEL VIA COMPACT-RUNTIME

0

SALARIES ON CHAIN · EVER

```
deployed at a42b898d...44c8 (block 11)
enrolled alice — commitment 727fd19c... tx 004fd009... (block 15)
enrolled bob · carol · dan — each with a proof from the proof server
attested alice · bob · carol · dan (4 txs)
publishReport proven & finalized tx 00826901... (block 54)
on-chain PayReport: women 2 · men 2 · mean gap 23.52% · salaries on chain: NONE
```

Omit an employee. Claim a zero gap. The same assertions reject both.

PRIVATE PAYROLL • NEVER LEAVES THIS PANEL

8 records

A. Serrano

ENGINEER • W

€ 62000

2ff9f2...8bd4

B. Keller

ENGINEER • M

€ 71000

5d60c9...f902

C. Okafor

DESIGNER • W

€ 55000

2ab431...bf6b

D. Novak

SALES • M

€ 68000

25384f...477b

E. Laurent

ENG. MANAGER • W

€ 90000

459f3e...e5c0

F. Braun

PRODUCT LEAD • M

€ 105000

1850e1...a7ab

G. Diallo

ANALYST • W

€ 58000

9dcb05...746e

H. Vidal

ANALYST • M

€ 64000

eede5f...1c81

PUBLIC LEDGER • WHAT THE CHAIN SEES

1 • enroll

2 • attest

3 • publish + prove

commitments (Merkle leaves)

8

nullifiers

8

employer attestations

8

salaries visible on-chain

0

ADVERSARIAL TOGGLES

☒ Omit one employee from the witness

☐ Claim the gap is 0.00%

PUBLISH REPORT

RESET

CIRCUIT REJECTED • assert failed: payroll witness must cover every enrolled employee (7 of 8)

Honest about what a proof can prove.

IT DOES NOT CLAIM	WHAT IS TRUE INSTEAD
Salaries hidden from HR	HR already holds payroll. Hidden from the public, colleagues, the regulator and Equilux. Wave 3: payroll provider attests.
The Article 10 trigger	Our flag is the company-wide mean $\geq 5\%$, named for what it measures. Article 10 is per worker category — Wave 2.
The full Article 9 return	One of seven metrics: the mean. Median, variable pay, quartiles follow.
Production scale	A sized instance: 32 commitment slots, 16-record witness. Sizing is a compile-time parameter.
A model of identity	Binary gender markers, as the Directive's reporting annex is written today.

Built in the open, wave by wave.

Wave 1 · delivered

Prove it works

Compact contract, 4 circuits with keys · 21 tests on the compiled circuits · end-to-end run on a Midnight network with real proofs · interactive demo with adversarial toggles · honest scope.

Wave 2

The product

Public testnet deployment with the site reading live state · Article 7 cohort-safe category query (k-anonymity) · worker categories · median · employer / employee / regulator screens · Lace wallet writes.

Wave 3

Fundable

Payroll-provider attestation (Personio / DATEV mock) · per-category Article 10 trigger · Article 9-shaped filing pack with proof hash + verify URL · instance sized past 250 · one named pilot.

The notary for Directive 2023/970.

Buyers

Works councils that don't trust the numbers. GDPR-sensitive employers. Regulators who want completeness without holding payroll.

Not competing with

Workday, SAP, Sysarb, Figures. They compute pay. We prove the set was complete and the figure matches — next to them, not instead of them.

Why Midnight

A forced legal deadline plus a problem only dual-ledger selective disclosure solves honestly. Incumbents can copy a dashboard; they cannot copy the proof.

